

RELATORIO DE ANALISE - MODELO VARMAX

Hiperparametros do Modelo:

```
p      (endog_lag)           : 1
x_lag (exog_lag)           : 1
r      (exog_real_degree)   : 1
i      (exog_img_degree)    : 2
```

Equacao do Modelo:

$$Y(t) = AR(1..1) + X_{real}^{(1..1)} + X_{img}^{(1..2)} \text{ com lag}(1..1)$$

Base de Dados:

```
Tamanho           : 48383
N° Replicações    : 1
Randomizada       : False
Seed              : 42
```

Resumo Estatístico:

Statespace Model Results

Dep. Variable:	['Yreal', 'Yimg']	No. Observations:	290298
Model:	VARX(1)	Log Likelihood	-591295.944
	+ intercept	AIC	1182633.888
Date:	Sun, 10 May 2026	BIC	1182856.040
Time:	09:42:18	HQIC	1182698.232
Sample:	0		
	- 290298		
Covariance Type:	opg		

METRICAS DO MODELO

Hiperparametros: p=1, x_lag=1, r=1, i=2

Metricas Gerais:

RMSE Medio	: 0.669177
AIC	: 1182633.8877
BIC	: 1182856.0396

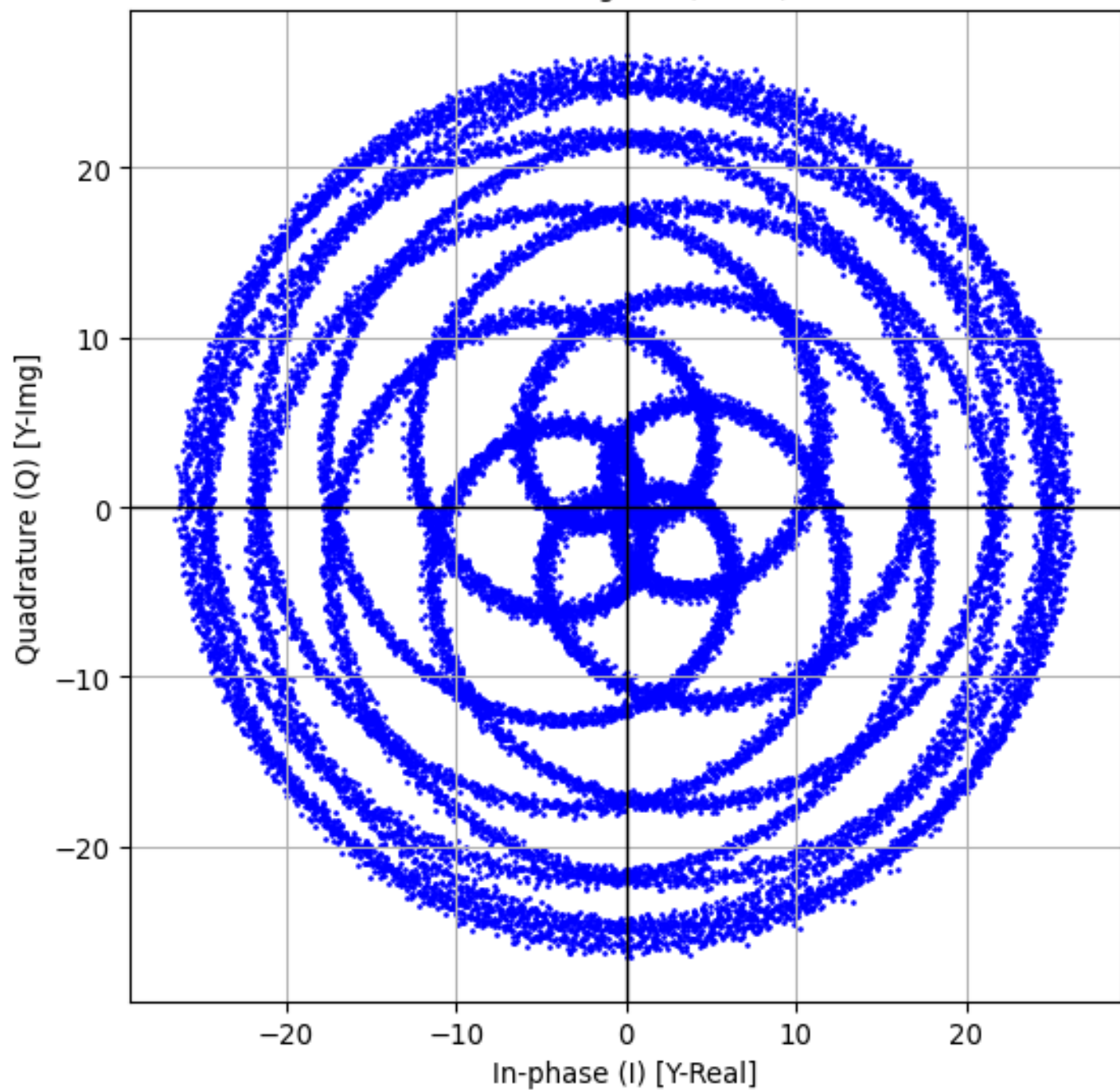
Componente Real (Yreal):

MSE	: 0.447391
RMSE	: 0.668873
MAE	: 0.515350
R2	: 0.997364
Adj R2	: 0.997364
STD	: 0.668872

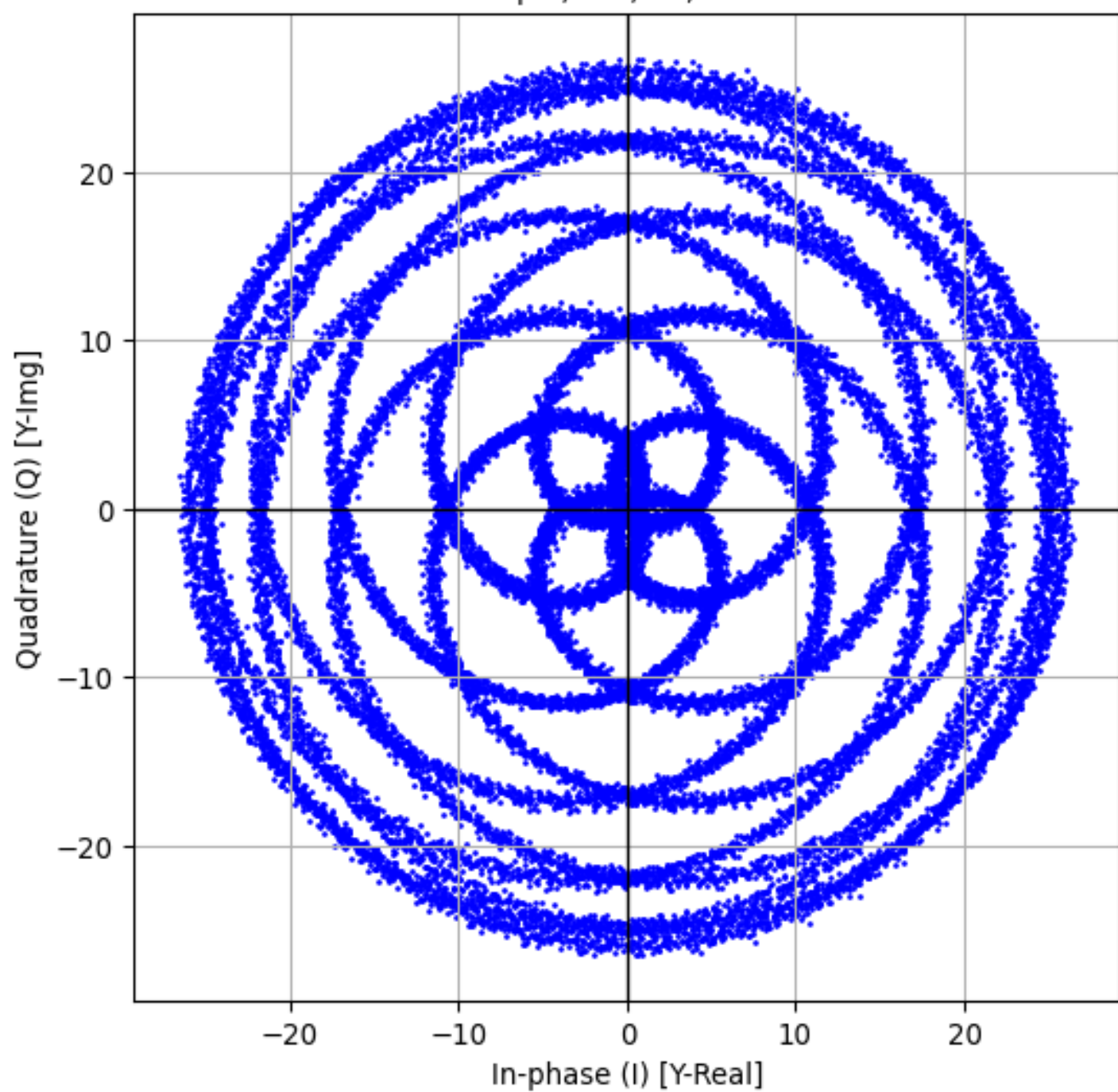
Componente Imaginaria (Yimg):

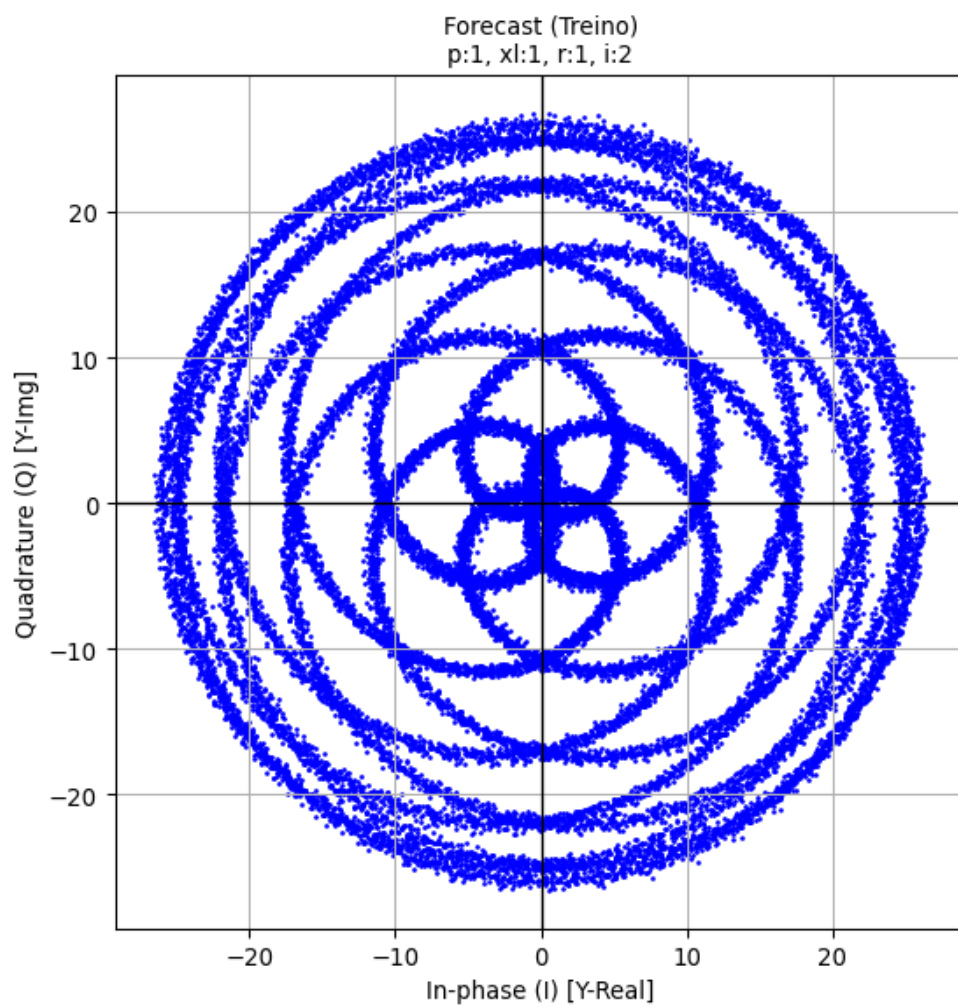
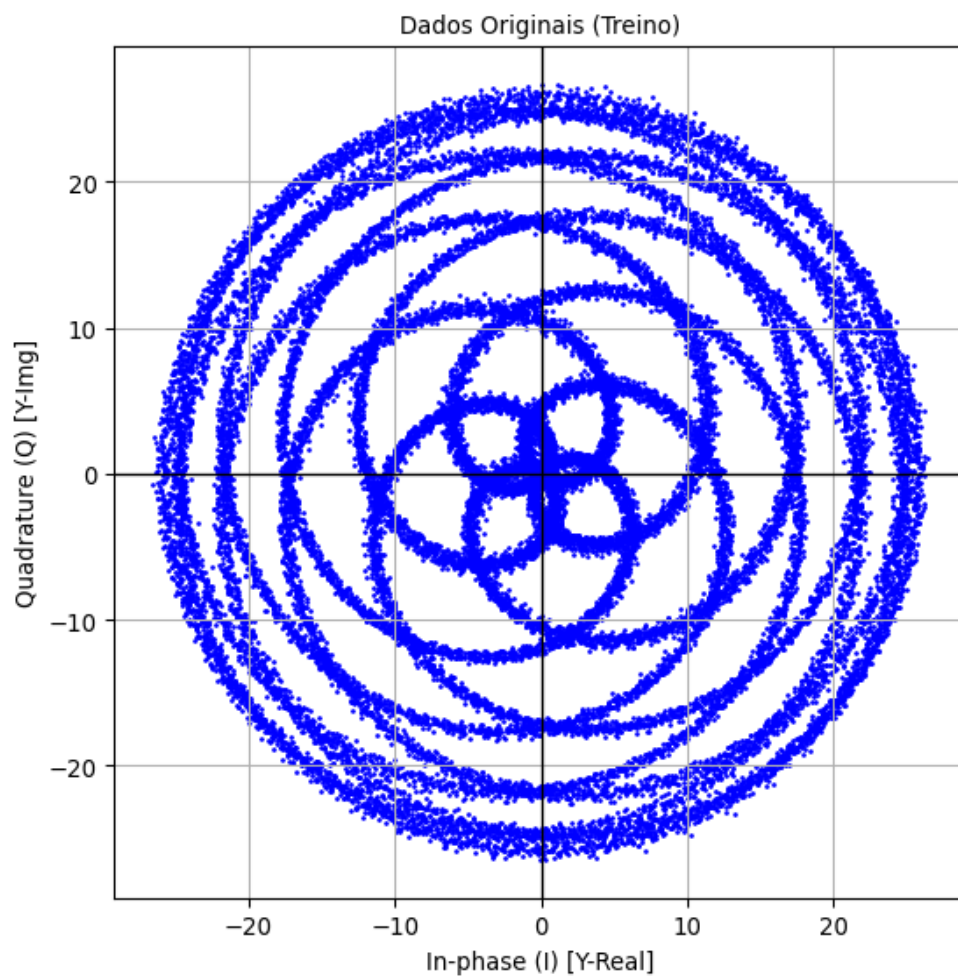
MSE	: 0.448206
RMSE	: 0.669482
MAE	: 0.515328
R2	: 0.997352
Adj R2	: 0.997352
STD	: 0.669482

Dados Originais (Treino)

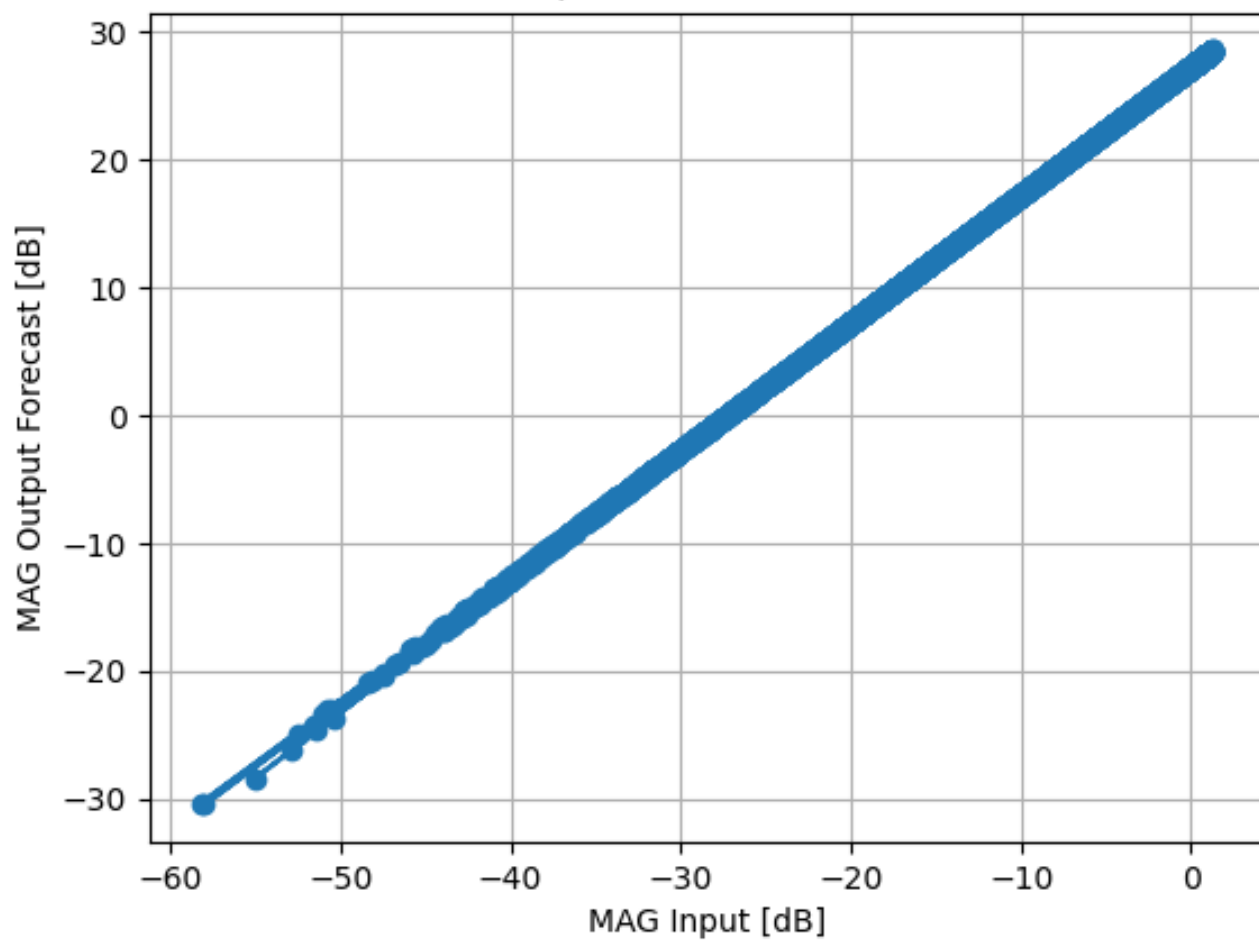


Forecast (Treino)
p:1, xl:1, r:1, i:2

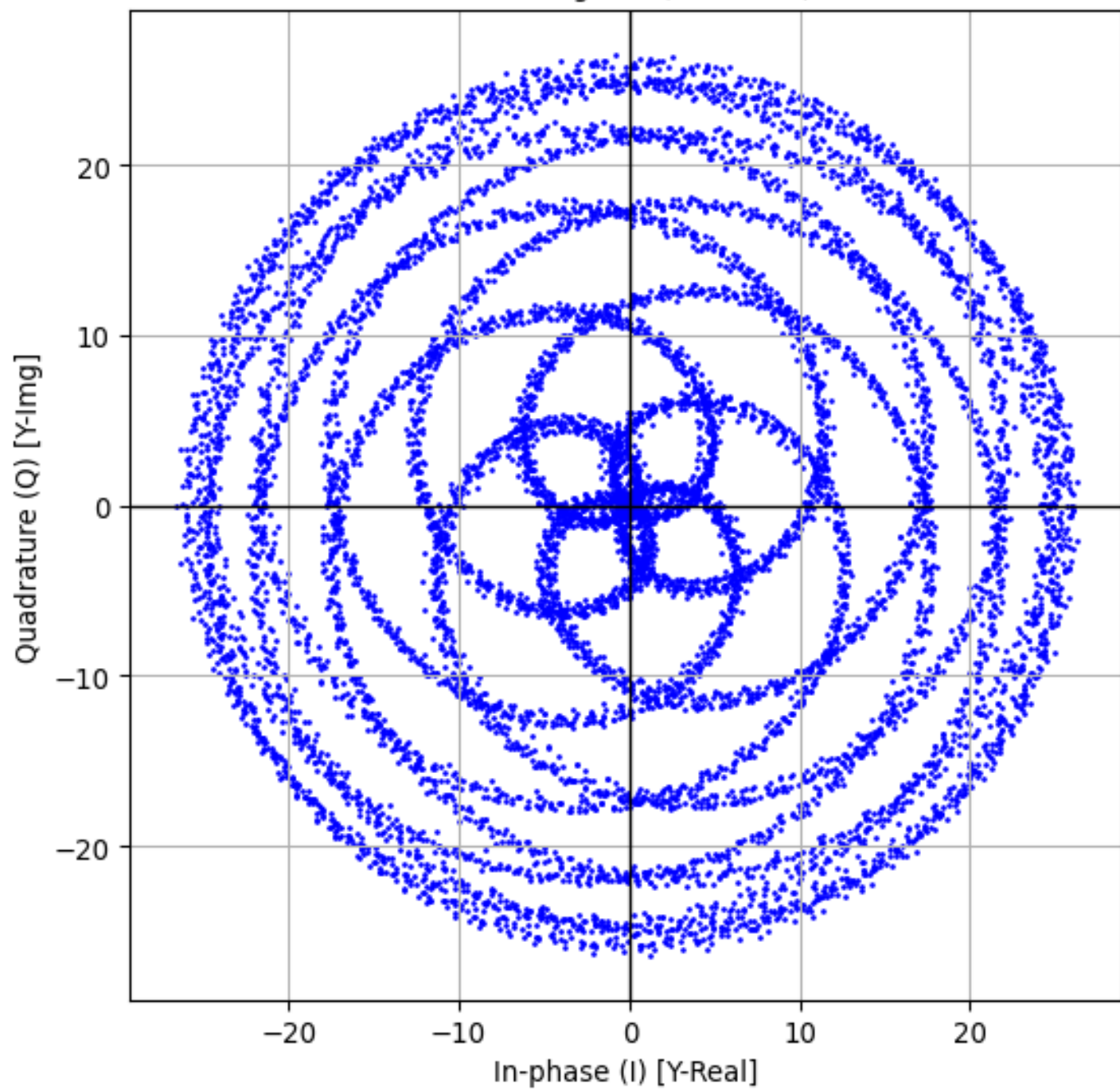




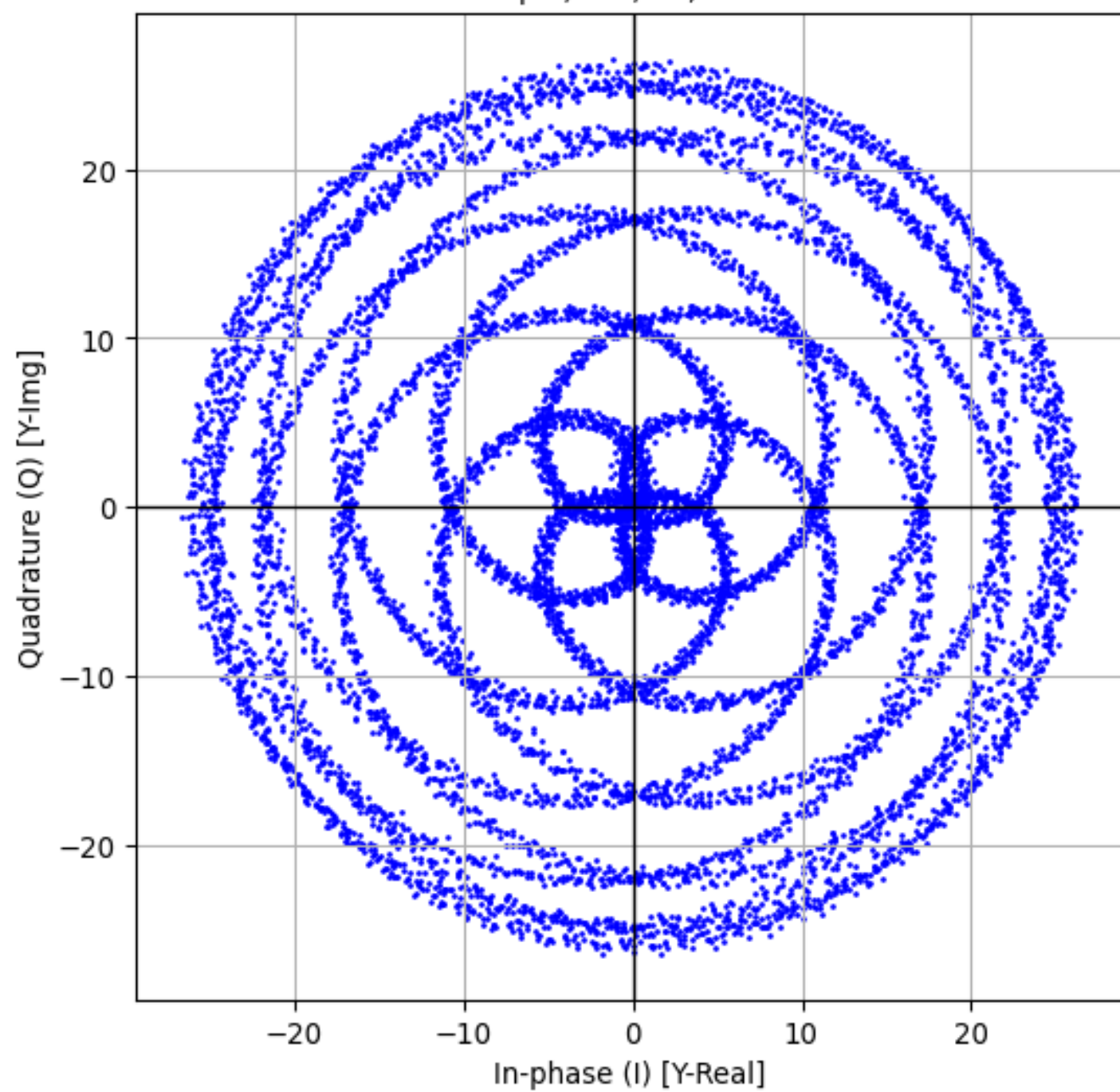
Mapa de Magnitude (Treino)
p:1, xl:1, r:1, i:2

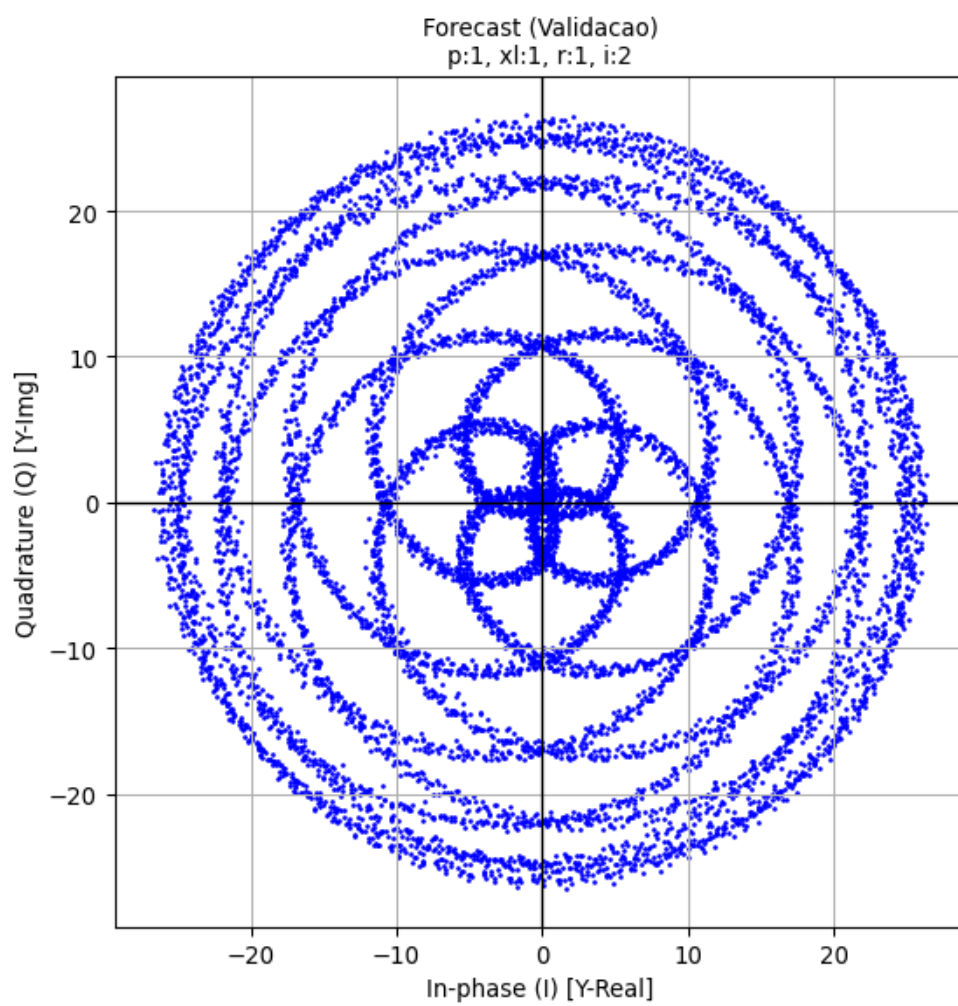
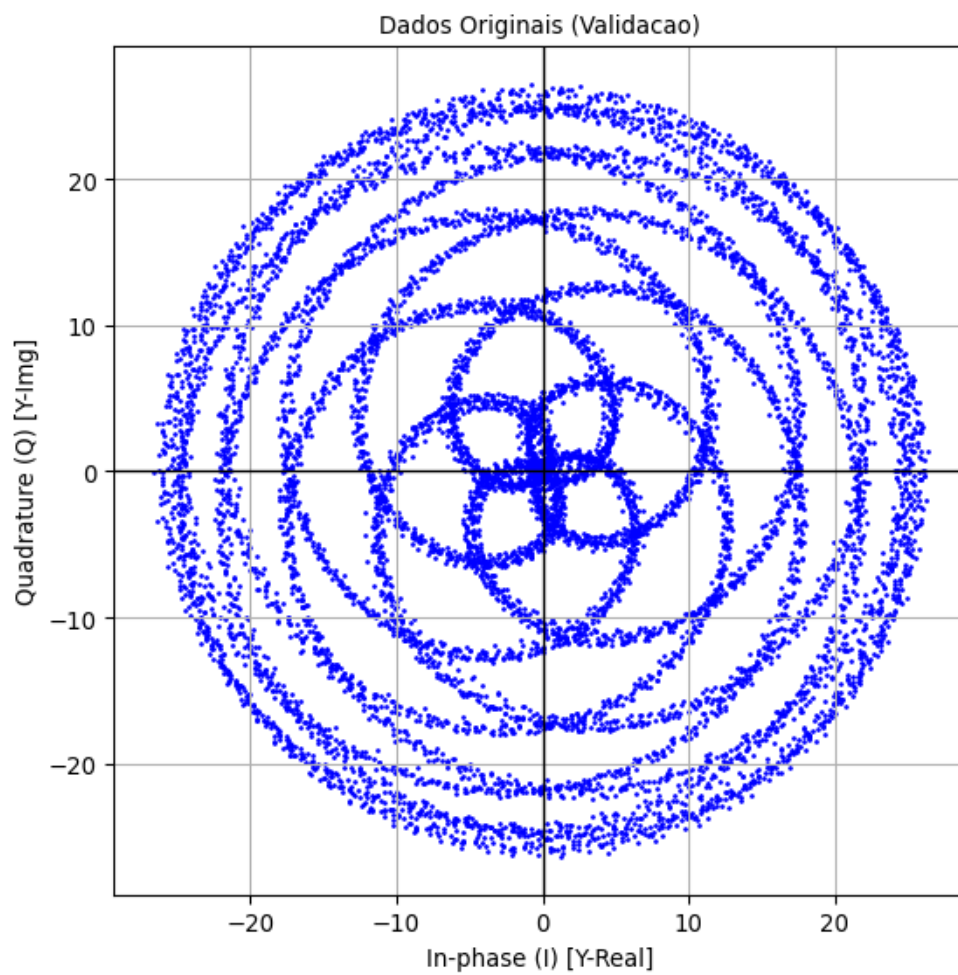


Dados Originais (Validacao)

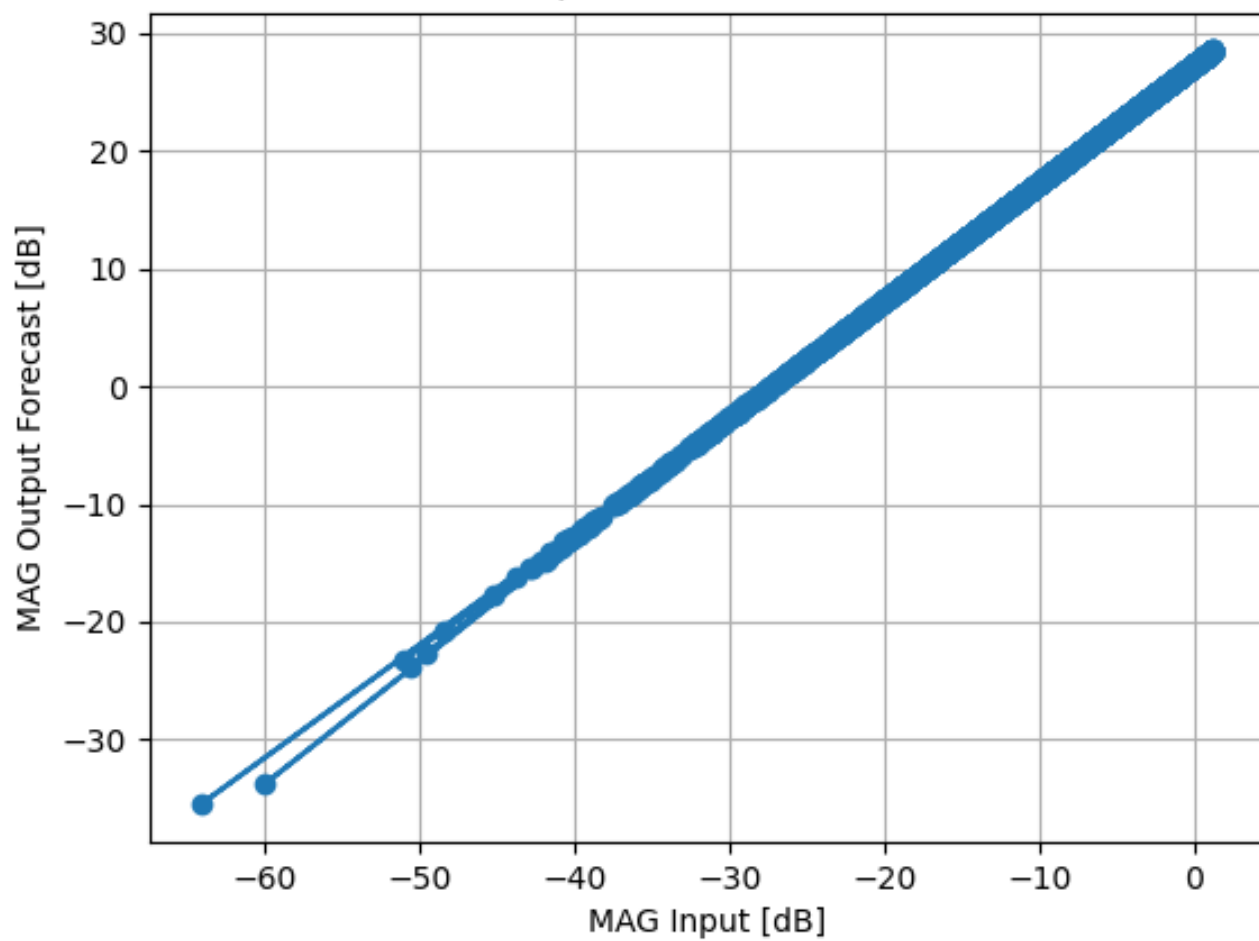


Forecast (Validacao)
p:1, xl:1, r:1, i:2

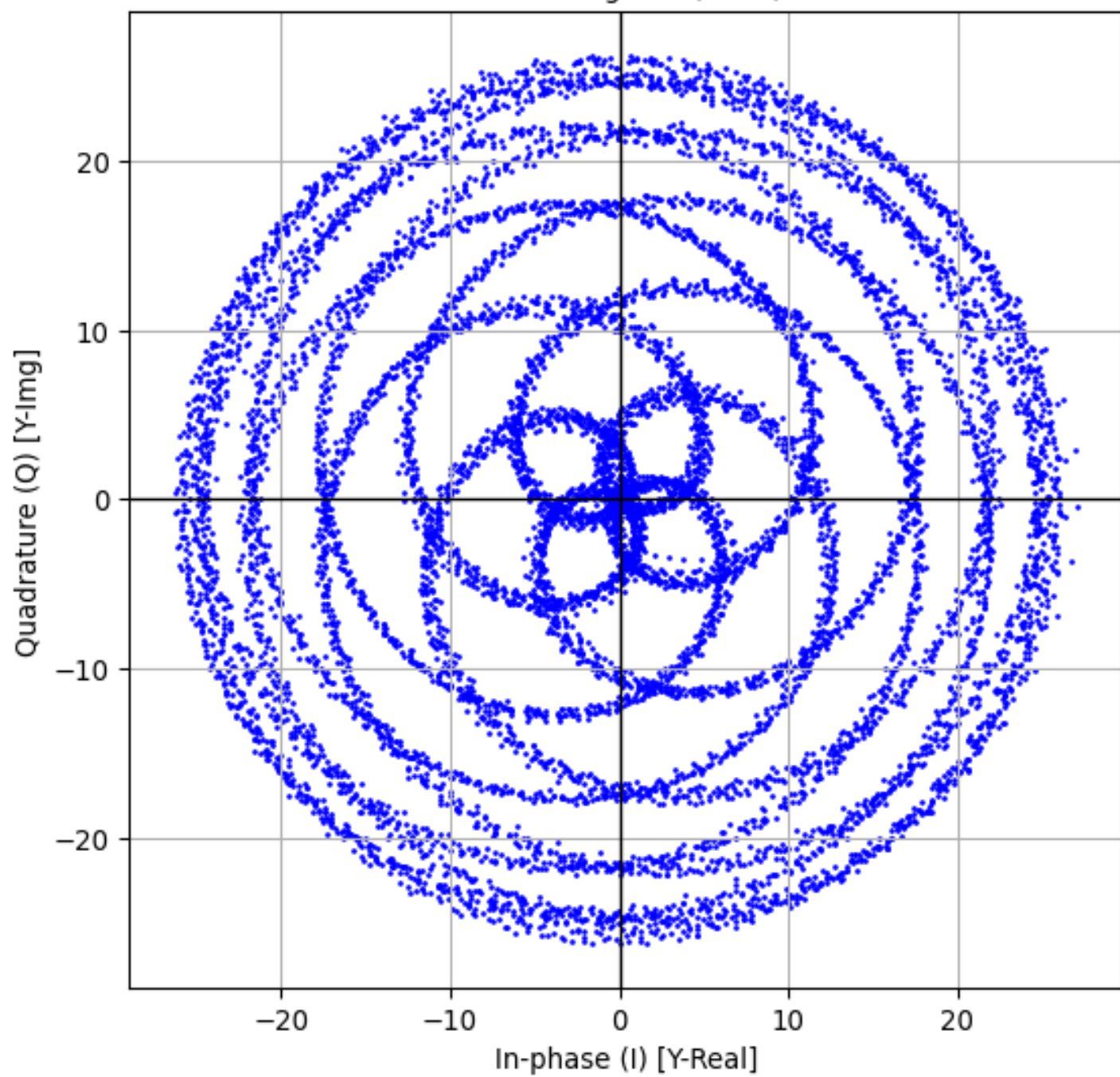




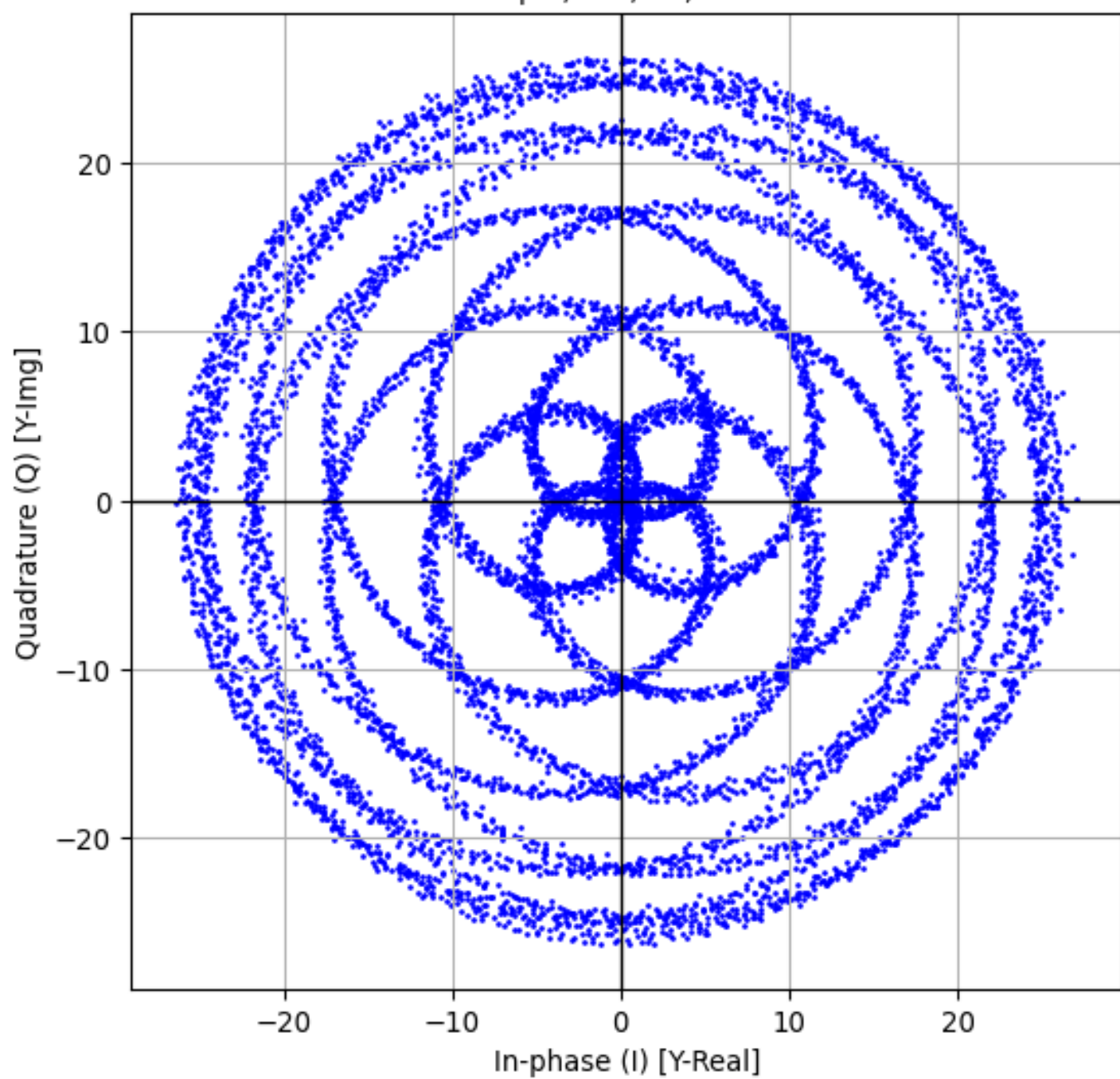
Mapa de Magnitude (Validacao)
p:1, xl:1, r:1, i:2

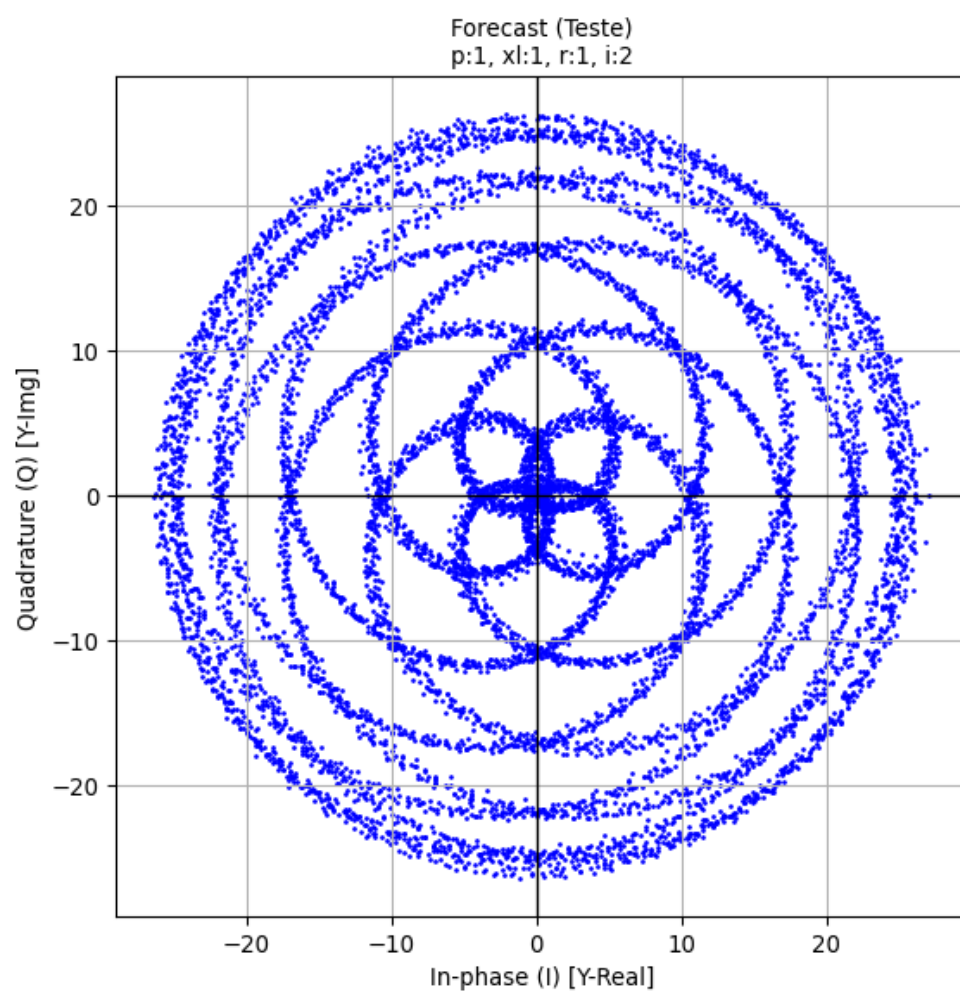
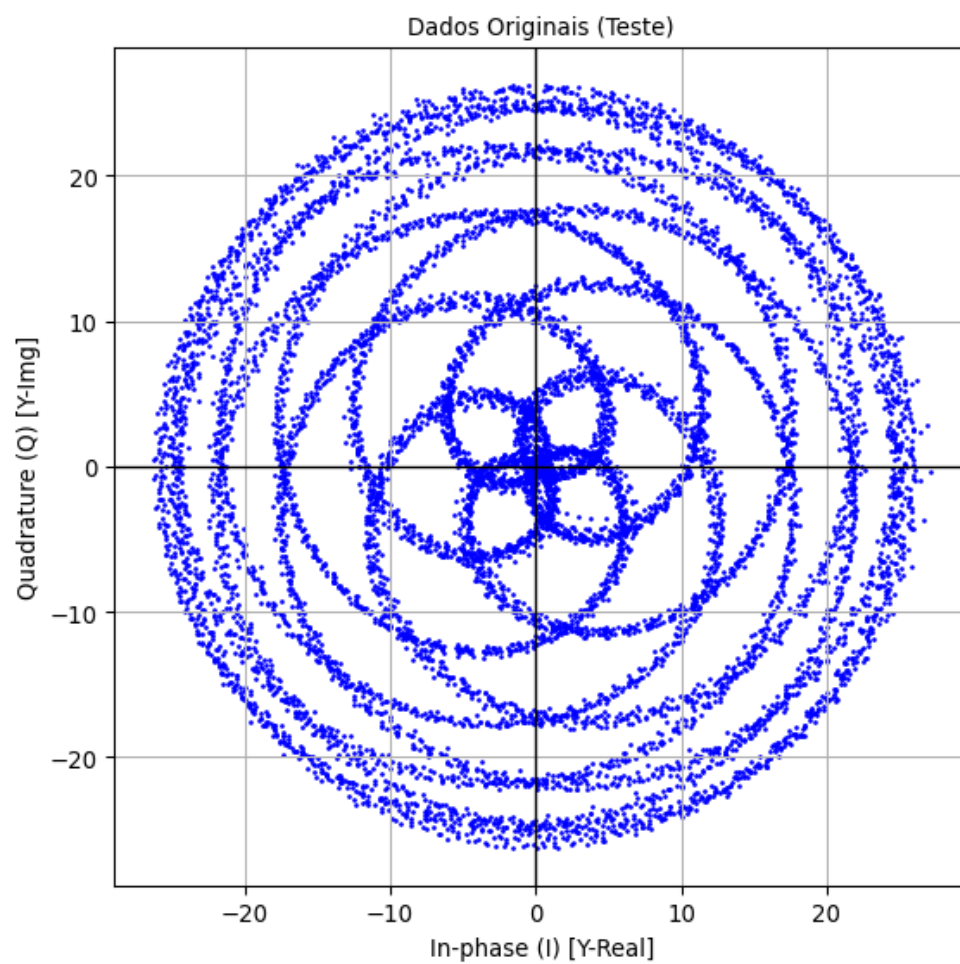


Dados Originais (Teste)

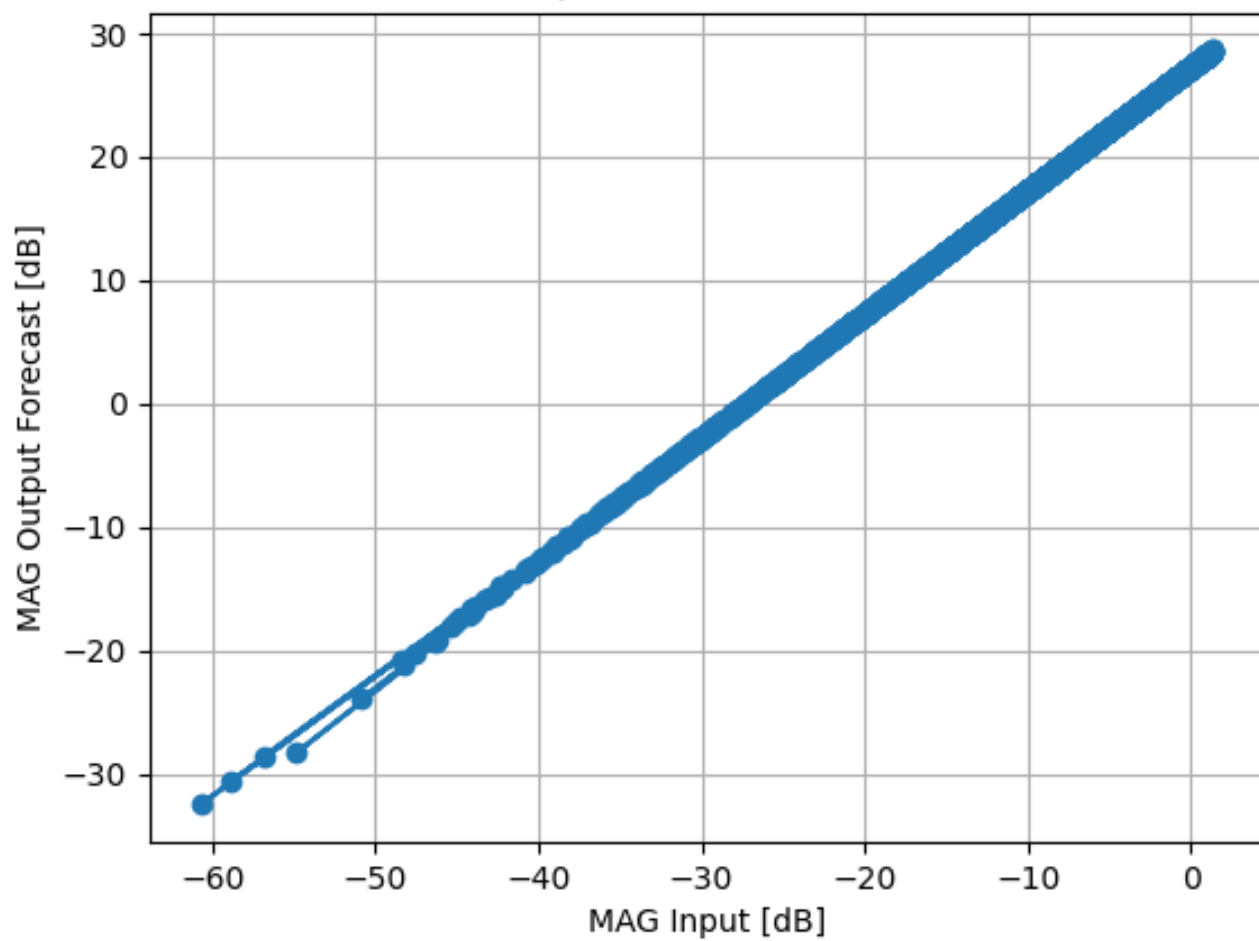


Forecast (Teste)
p:1, xl:1, r:1, i:2

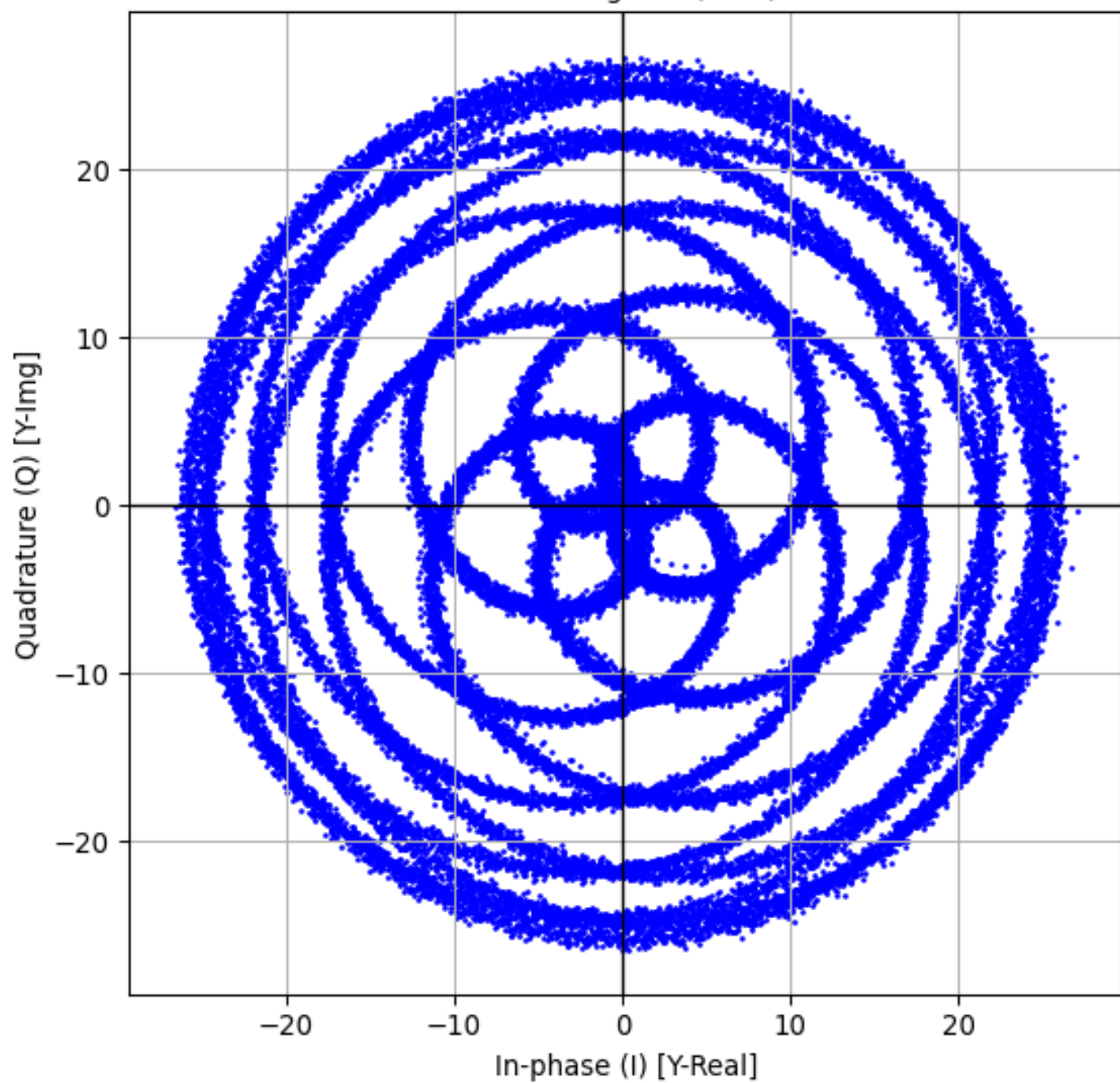




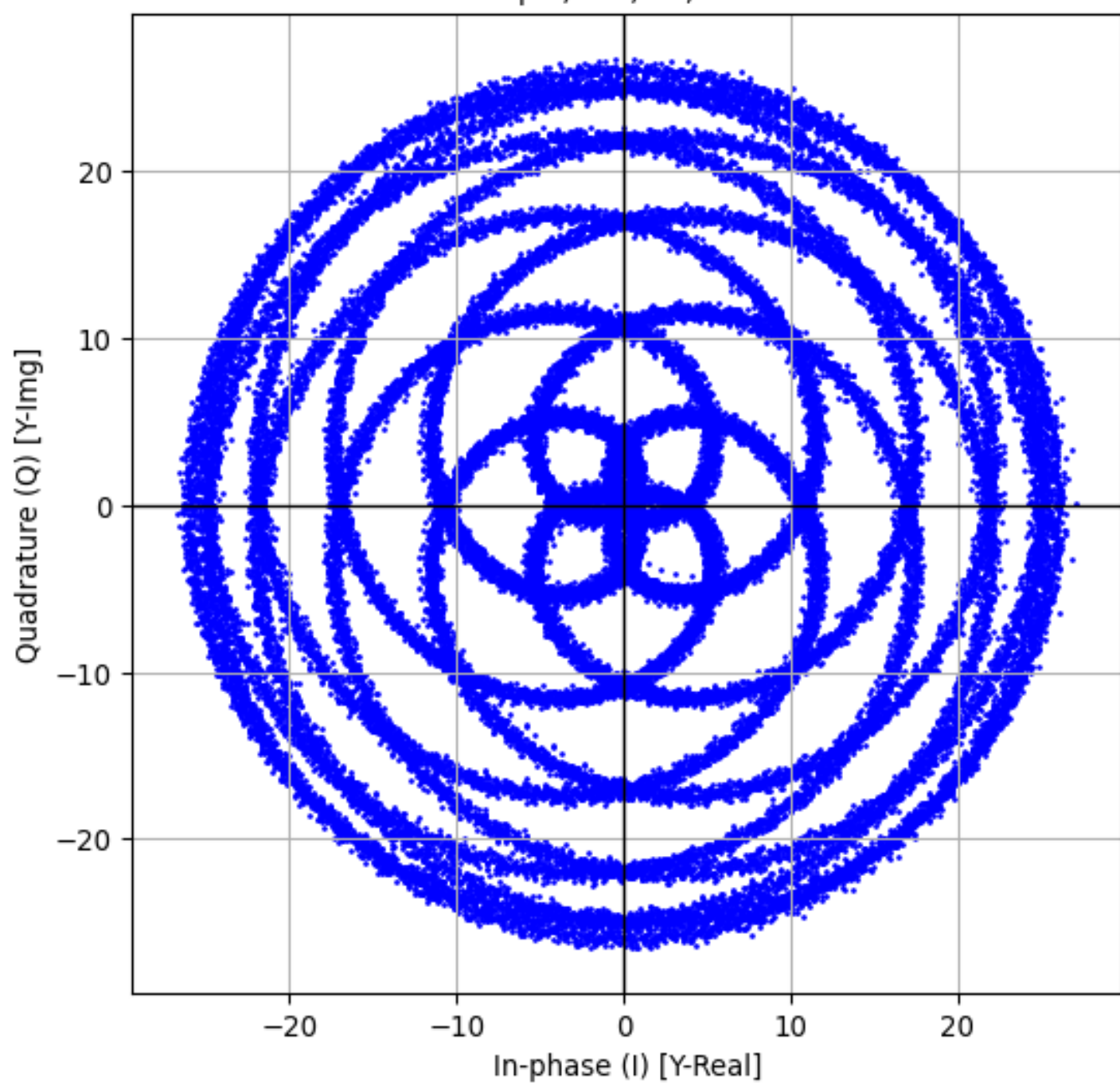
Mapa de Magnitude (Teste)
p:1, xl:1, r:1, i:2

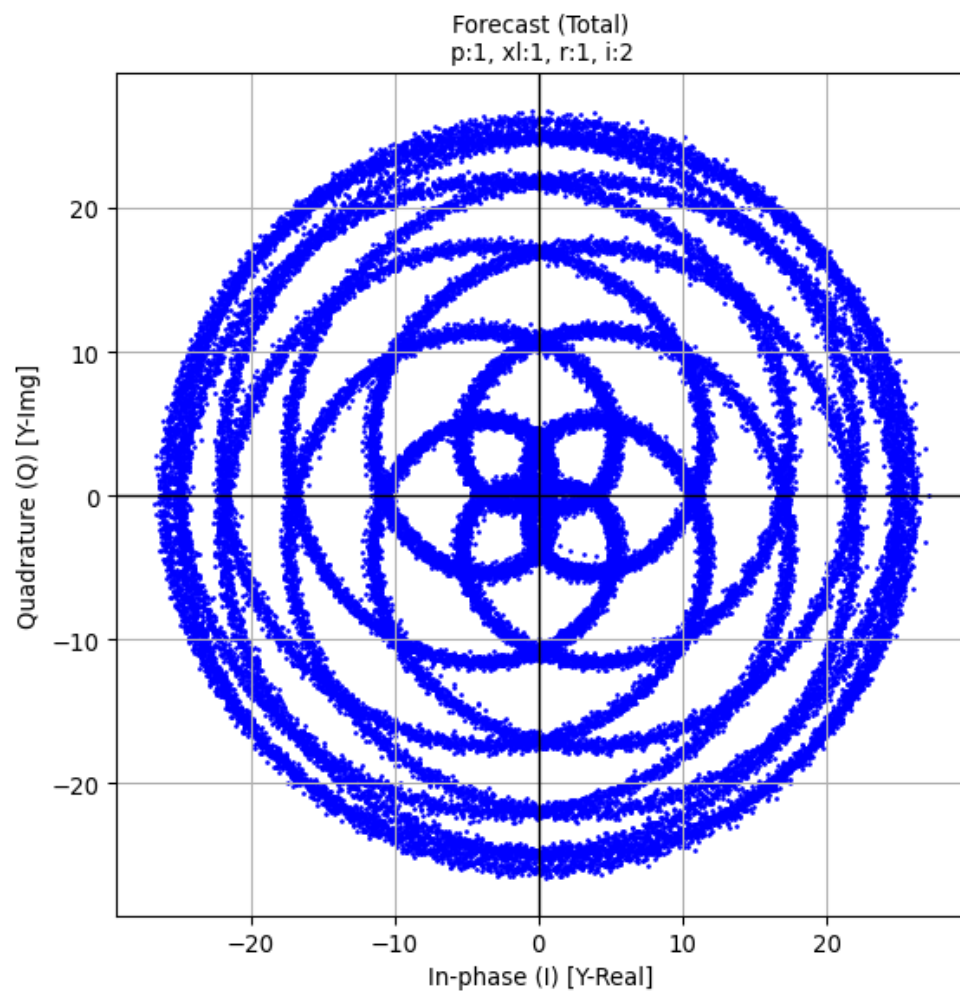
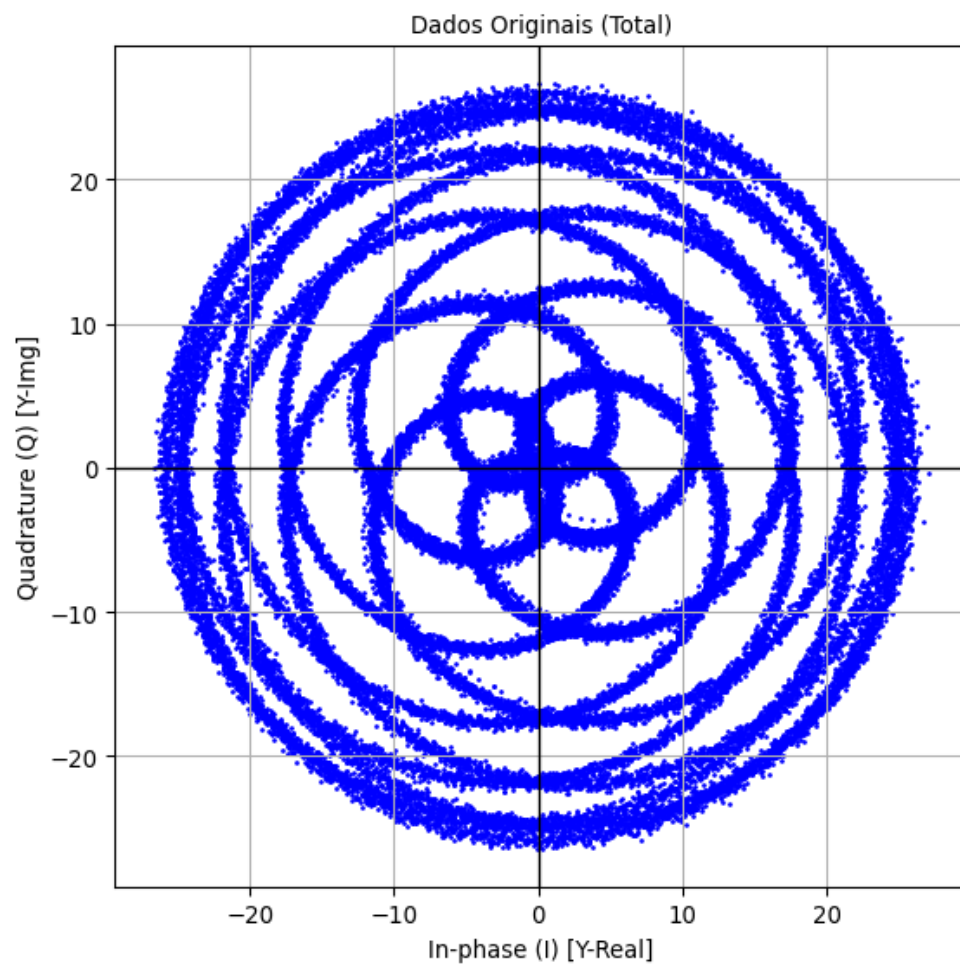


Dados Originais (Total)



Forecast (Total)
p:1, xl:1, r:1, i:2





Mapa de Magnitude (Total)
p:1, xl:1, r:1, i:2

